



Shanto-Mariam University of Creative Technology

Software Requirements Specification

for

Project: Falcon – Service Marketplace Platform

Version 1.0

By

Amina
Shahajadi Akter

Prepared for

Tauhid Islam

Department of CSE & CSIT

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the functional and non-functional requirements of **Falcon**, a service marketplace platform where users can both **offer** and **hire services** (local or remote).

This document is intended for:

- Developers
- Designers
- Product managers
- Stakeholders

1.2 Scope

Falcon is a mobile application (Flutter-based) that allows:

- Service providers to create profiles showcasing their skills, portfolio, and reviews
- Service receivers to search, compare, and hire providers
- Communication between users
- Booking and managing services

Services can be:

- **Local (e.g., electrician, cleaner, tutor)**
- **Remote (e.g., designer, developer, writer)**

1.3 Definitions

Term	Meaning
Provider	User offering services
Receiver	User hiring services
Gig/Service	A specific offering by a provider
Booking	Hiring request made by a receiver

Portfolio	Provider's previous work
Review	Feedback given after service completion

2. Overall Description

2.1 Product Perspective

Falcon is a **two-sided marketplace platform** similar to:

- Fiverr (remote services)
- Urban Company (local services)

System components:

- Mobile app (Flutter)
- Backend API
- Database
- Cloud storage

2.2 User Classes

1. Providers

- Create profiles
- List services
- Manage bookings
- Communicate with clients

2. Receivers

- Search providers
- Hire services
- Leave reviews

3. Admin (Future)

- Manage users
- Moderate content
- Handle disputes

3. System Features

3.1 Authentication & User Management

Features:

- User registration (email / phone / social login)
- Login / logout
- Profile creation
- Role selection (Provider / Receiver / Both)

Requirements:

- Users must verify email or phone
- Secure authentication (JWT or Firebase)

3.2 Provider Profile

Features:

- Profile photo
- About section
- Skills/services list
- Portfolio (images, videos, links)
- Reviews and ratings
- Availability (optional)

Requirements:

Providers must complete profile before listing services

3.3 Service Listings (Gig System)

Features:

- Create service listing
- Category selection

- Pricing (fixed or hourly)
- Location (for local services)
- Tags/keywords

Requirements:

Each service must have:

- Title
- Description
- Price

3.4 Search & Discovery

Features:

- Search by keyword
- Filter by:
 - Location
 - Price
 - Rating
 - Category
- Sort results

Requirements:

- Fast search response
- Relevant suggestions

3.5 Booking System

Features:

- Request service
- Accept/reject booking (provider)
- Schedule service
- Track status

Booking States:

Pending → Accepted → In Progress → Completed → Reviewed

3.6 Chat System

Features:

- Real-time messaging
- File/image sharing
- Pre-booking discussion

Requirements:

- Instant message delivery
- Notification support

3.7 Reviews & Ratings

Features:

- Rate provider (1–5 stars)
- Write review
- View reviews

Requirements:

- Only allowed after completed booking

3.8 Notifications

Features:

- Booking updates
- Messages
- Reviews
- Promotions (optional)

3.9 Admin Panel (Future)

Features:

- User management
- Content moderation
- Dispute handling

4. External Interface Requirements

4.1 User Interface

- Clean and intuitive UI
- Mobile-first design
- Dark/light mode (optional)

4.2 Hardware Interface

- Smartphone (Android/iOS)
- Internet connection required

4.3 Software Interface

- Backend API (REST/GraphQL)
- Cloud storage (S3, Firebase)
- Database (PostgreSQL)

5. Non-Functional Requirements

5.1 Performance

- App should load within 2–3 seconds
- Search results under 1 second

5.2 Security

- Encrypted data (HTTPS)
- Secure authentication
- Protection against spam/fraud

5.3 Scalability

- Support thousands of users
- Scalable backend architecture

5.4 Reliability

- 99% uptime
- Minimal crashes

5.5 Usability

- Easy onboarding
- Simple navigation

6. System Architecture (High-Level)

Mobile App (Flutter)



Backend API



Database + Storage



Realtime Services (Chat/Notifications)

7. Database Overview

Main entities:

Users

Profiles

Services

Bookings

Messages

Reviews

Notifications

8. Constraints

- Requires internet connection
- Dependent on backend uptime
- Legal considerations for payments/services

9. Future Enhancements

- AI-based provider recommendation
- Smart pricing suggestions
- Video consultations
- Subscription plans
- Location tracking for local services

10. Assumptions

- Users have smartphones
- Providers are responsible for service quality
- Platform acts as a mediator

11. Risks

- Fake profiles
- Payment fraud
- Poor service quality
- Scalability issues

12. Conclusion

Falcon aims to become a **hybrid service marketplace** combining:

- Local services (like Urban Company)
- Remote freelancing (like Fiverr)

The platform will focus on:

- Trust (reviews & profiles)
- Ease of hiring
- Real-time communication